Diabetic Retinopathy Data Analysis

2026-11-05

1. Dataset

```
Retinopathy_Debrecen <- read.csv(
  "C:/Users/ZahraPc/Downloads/archive (2)/Retinopathy_Debrecen.csv",
  header = TRUE
)
```

2. Dataset Structure and Variables

```
str(Retinopathy_Debrecen)
```

```
## 'data.frame':    1151 obs. of  19 variables:
##  $ quality                  : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ pre_screening            : int  1 1 1 1 1 1 0 1 1 1 ...
##  $ ma1                      : int  22 24 62 55 44 44 29 6 22 79 ...
##  $ ma2                      : int  22 24 60 53 44 43 29 6 21 75 ...
##  $ ma3                      : int  22 22 59 53 44 41 29 6 18 73 ...
##  $ ma4                      : int  19 18 54 50 41 41 27 6 15 71 ...
##  $ ma5                      : int  18 16 47 43 39 37 25 2 13 64 ...
##  $ ma6                      : int  14 13 33 31 27 29 16 1 10 47 ...
##  $ exudate1                 : num  49.9 57.7 55.8 40.5 18 ...
##  $ exudate2                 : num  17.78 23.8 27.99 18.45 8.57 ...
##  $ exudate3                 : num  5.27 3.33 12.69 9.12 0.41 ...
##  $ exudate5                 : num  0.0186 0.0039 1.3939 0.8403 0 ...
##  $ exudate6                 : num  0.00686 0.0039 0.37325 0.27243 0 ...
##  $ exudate7                 : num  0.00392 0.0039 0.04182 0.00765 0 ...
##  $ exudate8                 : num  0.00392 0.0039 0.00774 0.00153 0 ...
##  $ macula_opticdisc_distance: num  0.487 0.521 0.531 0.483 0.476 ...
##  $ opticdisc_diameter       : num  0.1 0.144 0.129 0.115 0.124 ...
##  $ am_fm_classification     : int  1 0 0 0 0 0 0 1 0 0 ...
##  $ class                    : int  0 0 1 0 1 1 1 0 1 1 ...
```

```
names(Retinopathy_Debrecen)
```

```
##  [1] "quality"                   "pre_screening"
##  [3] "ma1"                       "ma2"
##  [5] "ma3"                       "ma4"
##  [7] "ma5"                       "ma6"
##  [9] "exudate1"                  "exudate2"
```

```
## [11] "exudate3"                  "exudate5"
## [13] "exudate6"                  "exudate7"
## [15] "exudate8"                  "macula_opticdisc_distance"
## [17] "opticdisc_diameter"        "am_fm_classification"
## [19] "class"
```

```
head(Retinopathy_Debrecen, 15)
```

```
##    quality pre_screening ma1 ma2 ma3 ma4 ma5 ma6  exudate1  exudate2  exudate3
## 1        1             1  22  22  22  19  18  14 49.895756 17.775994  5.270920
## 2        1             1  24  24  22  18  16  13 57.709936 23.799994  3.325423
## 3        1             1  62  60  59  54  47  33 55.831441 27.993933 12.687485
## 4        1             1  55  53  53  50  43  31 40.467228 18.445954  9.118901
## 5        1             1  44  44  44  41  39  27 18.026254  8.570709  0.410381
## 6        1             1  44  43  41  41  37  29 28.356400  6.935636  2.305771
## 7        1             0  29  29  29  27  25  16 15.448398  9.113819  1.633493
## 8        1             1   6   6   6   6   2   1 20.679649  9.497786  1.223660
## 9        1             1  22  21  18  15  13  10 66.691933 23.545543  6.151117
## 10       1             1  79  75  73  71  64  47 22.141784 10.054384  0.874633
## 11       1             1  45  45  45  43  40  32 84.358401 50.977459 17.293722
## 12       1             0  25  25  25  23  22  18 22.480047 13.949995  0.436232
## 13       1             1  70  69  65  63  63  50 10.560100  3.108358  0.625511
## 14       1             1  48  43  39  32  27  18 23.012798  6.737583  2.403903
## 15       1             1  94  93  92  89  86  77  8.610822  1.981319  0.401183
##    exudate5 exudate6 exudate7 exudate8 macula_opticdisc_distance
## 1  0.018632 0.006864 0.003923 0.003923                 0.486903
## 2  0.003903 0.003903 0.003903 0.003903                 0.520908
## 3  1.393889 0.373252 0.041817 0.007744                 0.530904
## 4  0.840261 0.272434 0.007653 0.001531                 0.483284
## 5  0.000000 0.000000 0.000000 0.000000                 0.475935
## 6  0.000000 0.000000 0.000000 0.000000                 0.502831
## 7  0.000000 0.000000 0.000000 0.000000                 0.541743
## 8  0.000000 0.000000 0.000000 0.000000                 0.576318
## 9  0.000000 0.000000 0.000000 0.000000                 0.500073
## 10 0.023386 0.000000 0.000000 0.000000                 0.560959
## 11 0.000000 0.000000 0.000000 0.000000                 0.546008
## 12 0.000000 0.000000 0.000000 0.000000                 0.551682
## 13 0.103985 0.004799 0.000000 0.000000                 0.534396
## 14 0.011437 0.000000 0.000000 0.000000                 0.501554
## 15 0.000000 0.000000 0.000000 0.000000                 0.541277
##    opticdisc_diameter am_fm_classification class
## 1            0.100025                    1     0
## 2            0.144414                    0     0
## 3            0.128548                    0     1
## 4            0.114790                    0     0
## 5            0.123572                    0     1
## 6            0.126741                    0     1
## 7            0.139575                    0     1
## 8            0.071071                    1     0
## 9            0.116793                    0     1
## 10           0.109134                    0     1
## 11           0.112378                    0     0
## 12           0.139657                    1     0
## 13           0.089587                    0     1
```

```
## 14           0.138287                    1     1
## 15           0.124505                    0     0
```

3. User-Defined Function

```
retinopathy_status <- function(x) {
  ifelse(x == 1, "Signs Present", "No Signs")
}

Retinopathy_Debrecen$Status <- retinopathy_status(
  Retinopathy_Debrecen$class
)

table(Retinopathy_Debrecen$Status)
```

```
##
##      No Signs Signs Present
##           540           611
```

4. Filtering Data Using Logical Criteria

```
retinopathy_filtered <- subset(
  Retinopathy_Debrecen,
  class == 1 & quality == 1
)

head(retinopathy_filtered)
```

```
##    quality pre_screening ma1 ma2 ma3 ma4 ma5 ma6 exudate1  exudate2  exudate3
## 3        1             1  62  60  59  54  47  33 55.83144 27.993933 12.687485
## 5        1             1  44  44  44  41  39  27 18.02625  8.570709  0.410381
## 6        1             1  44  43  41  41  37  29 28.35640  6.935636  2.305771
## 7        1             0  29  29  29  27  25  16 15.44840  9.113819  1.633493
## 9        1             1  22  21  18  15  13  10 66.69193 23.545543  6.151117
## 10       1             1  79  75  73  71  64  47 22.14178 10.054384  0.874633
##    exudate5 exudate6 exudate7 exudate8 macula_opticdisc_distance
## 3  1.393889 0.373252 0.041817 0.007744                 0.530904
## 5  0.000000 0.000000 0.000000 0.000000                 0.475935
## 6  0.000000 0.000000 0.000000 0.000000                 0.502831
## 7  0.000000 0.000000 0.000000 0.000000                 0.541743
## 9  0.000000 0.000000 0.000000 0.000000                 0.500073
## 10 0.023386 0.000000 0.000000 0.000000                 0.560959
##    opticdisc_diameter am_fm_classification class        Status
## 3            0.128548                    0     1 Signs Present
## 5            0.123572                    0     1 Signs Present
## 6            0.126741                    0     1 Signs Present
## 7            0.139575                    0     1 Signs Present
## 9            0.116793                    0     1 Signs Present
## 10           0.109134                    0     1 Signs Present
```

```r
nrow(retinopathy_filtered)
```

```
## [1] 611
```

5. Dependent and Independent Variables

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)

Retinopathy_Debrecen$ID <- seq_len(nrow(Retinopathy_Debrecen))

dependent <- Retinopathy_Debrecen %>%
  select(ID, class)

independent <- Retinopathy_Debrecen %>%
  select(ID, ma1, ma2, ma3)
independent_long <- pivot_longer(
  independent,
  cols = c(ma1, ma2, ma3),
  names_to = "Measurement",
  values_to = "Value"
)

independent_wide <- pivot_wider(
  independent_long,
  names_from = Measurement,
  values_from = Value
)

combined_data <- inner_join(
  dependent,
  independent_wide,
  by = "ID"
)

head(combined_data)
```

```
##   ID class ma1 ma2 ma3
## 1  1     0  22  22  22
## 2  2     0  24  24  22
## 3  3     1  62  60  59
## 4  4     0  55  53  53
## 5  5     1  44  44  44
## 6  6     1  44  43  41
```

6. Removing Missing Values

```
# Count missing values in each column
colSums(is.na(Retinopathy_Debrecen))
```

```
##                   quality            pre_screening                      ma1
##                         0                        0                        0
##                       ma2                      ma3                      ma4
##                         0                        0                        0
##                       ma5                      ma6                  exudate1
##                         0                        0                        0
##                  exudate2                 exudate3                  exudate5
##                         0                        0                        0
##                  exudate6                 exudate7                  exudate8
##                         0                        0                        0
## macula_opticdisc_distance       opticdisc_diameter     am_fm_classification
##                         0                        0                        0
##                     class                   Status                       ID
##                         0                        0                        0
```

```
# Remove rows containing missing values
retinopathy_clean <- na.omit(Retinopathy_Debrecen)

# Compare rows before and after cleaning
nrow(Retinopathy_Debrecen)
```

```
## [1] 1151
```

```
nrow(retinopathy_clean)
```

```
## [1] 1151
```

7. Identifying and Removing Duplicate Data

```
# Check duplicates using the original dataset columns
original_columns <- setdiff(
  names(retinopathy_clean), c("ID", "Status")
)

# Count duplicate records
sum(duplicated(retinopathy_clean[original_columns]))
```

```
## [1] 5
```

```
# Remove duplicate records
retinopathy_clean <- retinopathy_clean[
  !duplicated(retinopathy_clean[original_columns]), ]

# Display remaining records
nrow(retinopathy_clean)
```

```
## [1] 1146
```

8. Reordering Rows in Descending Order

```
# Sort retinal measurements in descending order
retinopathy_sorted <- retinopathy_clean %>%
  arrange(desc(ma1), desc(ma2))

# Display the first 10 sorted records
head(retinopathy_sorted, 10)
```

```
##    quality pre_screening ma1 ma2 ma3 ma4 ma5 ma6  exudate1 exudate2  exudate3
## 1        1             1 151 131 120 105  84  57  45.00382 21.60080 13.209197
## 2        1             1 147 132 113  91  65  37  32.69369 10.34998  0.917888
## 3        1             1 141 130 110  84  62  34  88.46352 44.14061 20.072978
## 4        1             1 137 117 107  94  82  54  57.81878 17.03802  8.985278
## 5        1             1 126 114  98  75  62  40  22.74796 11.10276  3.432302
## 6        1             1 114 110  93  76  51  27 189.37048 27.80744 13.990668
## 7        1             1 114  99  94  84  76  49  31.64883 13.18841  2.217076
## 8        1             1 107  98  77  63  47  28 199.42903 44.82088 25.534925
## 9        1             1 107  97  89  73  61  39  46.89829 26.62532 11.290695
## 10       1             1 105  99  90  81  74  40  34.35939 18.13080  4.048434
##    exudate5 exudate6 exudate7 exudate8 macula_opticdisc_distance
## 1  3.618145 1.756558 0.715409 0.136993                 0.502597
## 2  0.000000 0.000000 0.000000 0.000000                 0.588851
## 3  4.467041 2.521083 1.708253 0.751390                 0.524378
## 4  2.128214 0.940230 0.655545 0.397020                 0.535003
## 5  0.796506 0.592314 0.483203 0.324214                 0.537198
## 6  0.008232 0.000000 0.000000 0.000000                 0.577045
## 7  0.021289 0.013686 0.013686 0.012165                 0.520957
## 8  0.242466 0.011447 0.003122 0.002081                 0.491310
## 9  5.258325 3.758994 2.169764 1.330199                 0.511867
## 10 0.283436 0.100036 0.040924 0.030314                 0.543699
##    opticdisc_diameter am_fm_classification class        Status   ID
## 1            0.114161                    0     1 Signs Present  340
## 2            0.085459                    0     1 Signs Present 1078
## 3            0.096694                    0     1 Signs Present  576
## 4            0.124646                    0     1 Signs Present  333
## 5            0.149637                    0     1 Signs Present  281
## 6            0.131706                    1     1 Signs Present  130
## 7            0.097320                    0     1 Signs Present  561
## 8            0.138403                    1     1 Signs Present   33
```

```
## 9            0.097517                    0     1 Signs Present   56
## 10           0.115193                    0     1 Signs Present  851
```

9. Renaming Column Names

```r
# Rename selected columns
retinopathy_clean <- retinopathy_clean %>%
  rename(
    Image_Quality = quality,
    Pre_Screening = pre_screening,
    Retinopathy_Outcome = class
  )

# Display updated column names
names(retinopathy_clean)
```

```
##  [1] "Image_Quality"             "Pre_Screening"
##  [3] "ma1"                       "ma2"
##  [5] "ma3"                       "ma4"
##  [7] "ma5"                       "ma6"
##  [9] "exudate1"                  "exudate2"
## [11] "exudate3"                  "exudate5"
## [13] "exudate6"                  "exudate7"
## [15] "exudate8"                  "macula_opticdisc_distance"
## [17] "opticdisc_diameter"        "am_fm_classification"
## [19] "Retinopathy_Outcome"       "Status"
## [21] "ID"
```

10. Adding New Variables Using Mathematical Functions

```r
# Create new variables using mathematical calculations
retinopathy_clean <- retinopathy_clean %>%
  mutate(
    ma1_Doubled = ma1 * 2,
    ma2_Average = (ma1 + ma2) / 2
  )

# Display the original and calculated variables
head(retinopathy_clean[, c("ma1", "ma2",
                           "ma1_Doubled", "ma2_Average")])
```

```
##   ma1 ma2 ma1_Doubled ma2_Average
## 1  22  22          44        22.0
## 2  24  24          48        24.0
## 3  62  60         124        61.0
## 4  55  53         110        54.0
## 5  44  44          88        44.0
## 6  44  43          88        43.5
```

11. Creating a Training Set

```
# Set a seed for reproducible random results
set.seed(123)

# Randomly select 70% of the records for training
train_index <- sample(
  seq_len(nrow(retinopathy_clean)),
  size = floor(0.7 * nrow(retinopathy_clean))
)

# Create training and testing datasets
train_set <- retinopathy_clean[train_index, ]
test_set <- retinopathy_clean[-train_index, ]

# Display dataset dimensions
dim(train_set)
```

```
## [1] 802  23
```

```
dim(test_set)
```

```
## [1] 344  23
```

12. Summary Statistics

```
# Display summary statistics for the cleaned dataset
summary(retinopathy_clean)
```

```
##  Image_Quality    Pre_Screening        ma1              ma2
##  Min.   :0.0000   Min.   :0.000   Min.   :  1.00   Min.   :  1.00
##  1st Qu.:1.0000   1st Qu.:1.000   1st Qu.: 16.00   1st Qu.: 16.00
##  Median :1.0000   Median :1.000   Median : 35.00   Median : 34.00
##  Mean   :0.9965   Mean   :0.918   Mean   : 38.37   Mean   : 36.85
##  3rd Qu.:1.0000   3rd Qu.:1.000   3rd Qu.: 55.00   3rd Qu.: 53.00
##  Max.   :1.0000   Max.   :1.000   Max.   :151.00   Max.   :132.00
##       ma3              ma4              ma5              ma6
##  Min.   :  1.00   Min.   :  1.00   Min.   : 1.00   Min.   : 1.00
##  1st Qu.: 15.00   1st Qu.: 14.00   1st Qu.:11.00   1st Qu.: 8.00
##  Median : 32.00   Median : 29.00   Median :25.00   Median :18.00
##  Mean   : 35.08   Mean   : 32.23   Mean   :28.67   Mean   :21.09
##  3rd Qu.: 51.00   3rd Qu.: 48.00   3rd Qu.:43.00   3rd Qu.:32.00
##  Max.   :120.00   Max.   :105.00   Max.   :97.00   Max.   :89.00
##     exudate1          exudate2         exudate3         exudate5
##  Min.   :  0.3493   Min.   :  0.000   Min.   :  0.000   Min.   : 0.00000
##  1st Qu.: 22.3362   1st Qu.:  7.944   1st Qu.:  1.270   1st Qu.: 0.00000
##  Median : 44.3920   Median : 17.061   Median :  4.446   Median : 0.02215
##  Mean   : 64.2611   Mean   : 23.135   Mean   :  8.727   Mean   : 0.56252
##  3rd Qu.: 87.9153   3rd Qu.: 31.311   3rd Qu.: 11.776   3rd Qu.: 0.19236
```

```
##  Max.   :403.9391   Max.   :167.131   Max.   :106.070   Max.   :51.42321
##     exudate6            exudate7           exudate8
##  Min.   : 0.000000   Min.   :0.000000   Min.   :0.00000
##  1st Qu.: 0.000000   1st Qu.:0.000000   1st Qu.:0.00000
##  Median : 0.001551   Median :0.000000   Median :0.00000
##  Mean   : 0.213042   Mean   :0.085952   Mean   :0.03732
##  3rd Qu.: 0.037403   3rd Qu.:0.004809   3rd Qu.:0.00385
##  Max.   :20.098605   Max.   :5.937799   Max.   :3.08675
##  macula_opticdisc_distance opticdisc_diameter am_fm_classification
##  Min.   :0.3678            Min.   :0.05791    Min.   :0.0000
##  1st Qu.:0.5027            1st Qu.:0.09579    1st Qu.:0.0000
##  Median :0.5233            Median :0.10664    Median :0.0000
##  Mean   :0.5232            Mean   :0.10847    Mean   :0.3377
##  3rd Qu.:0.5437            3rd Qu.:0.11975    3rd Qu.:1.0000
##  Max.   :0.5922            Max.   :0.21920    Max.   :1.0000
##  Retinopathy_Outcome       Status           ID          ma1_Doubled
##  Min.   :0.0000      Length   :1146   Min.   :   1.0   Min.   :  2.00
##  1st Qu.:0.0000      N.unique :   2   1st Qu.: 288.2   1st Qu.: 32.00
##  Median :1.0000      N.blank  :   0   Median : 574.5   Median : 70.00
##  Mean   :0.5288      Min.nchar:   8   Mean   : 575.9   Mean   : 76.73
##  3rd Qu.:1.0000      Max.nchar:  13   3rd Qu.: 864.8   3rd Qu.:110.00
##  Max.   :1.0000                       Max.   :1151.0   Max.   :302.00
##   ma2_Average
##  Min.   :  1.00
##  1st Qu.: 16.00
##  Median : 35.00
##  Mean   : 37.61
##  3rd Qu.: 54.00
##  Max.   :141.00
```

13. Mean, Median, Mode and Range

```r
# Calculate mean and median
mean_ma1 <- mean(retinopathy_clean$ma1)
median_ma1 <- median(retinopathy_clean$ma1)

# Create a user-defined function to calculate mode
get_mode <- function(x) {
  values <- unique(x)
  values[which.max(tabulate(match(x, values)))]
}

# Calculate mode and range
mode_ma1 <- get_mode(retinopathy_clean$ma1)
range_ma1 <- range(retinopathy_clean$ma1)

# Display results
mean_ma1
```

```
## [1] 38.36649
```

```
median_ma1
```

```
## [1] 35
```

```
mode_ma1
```

```
## [1] 16
```

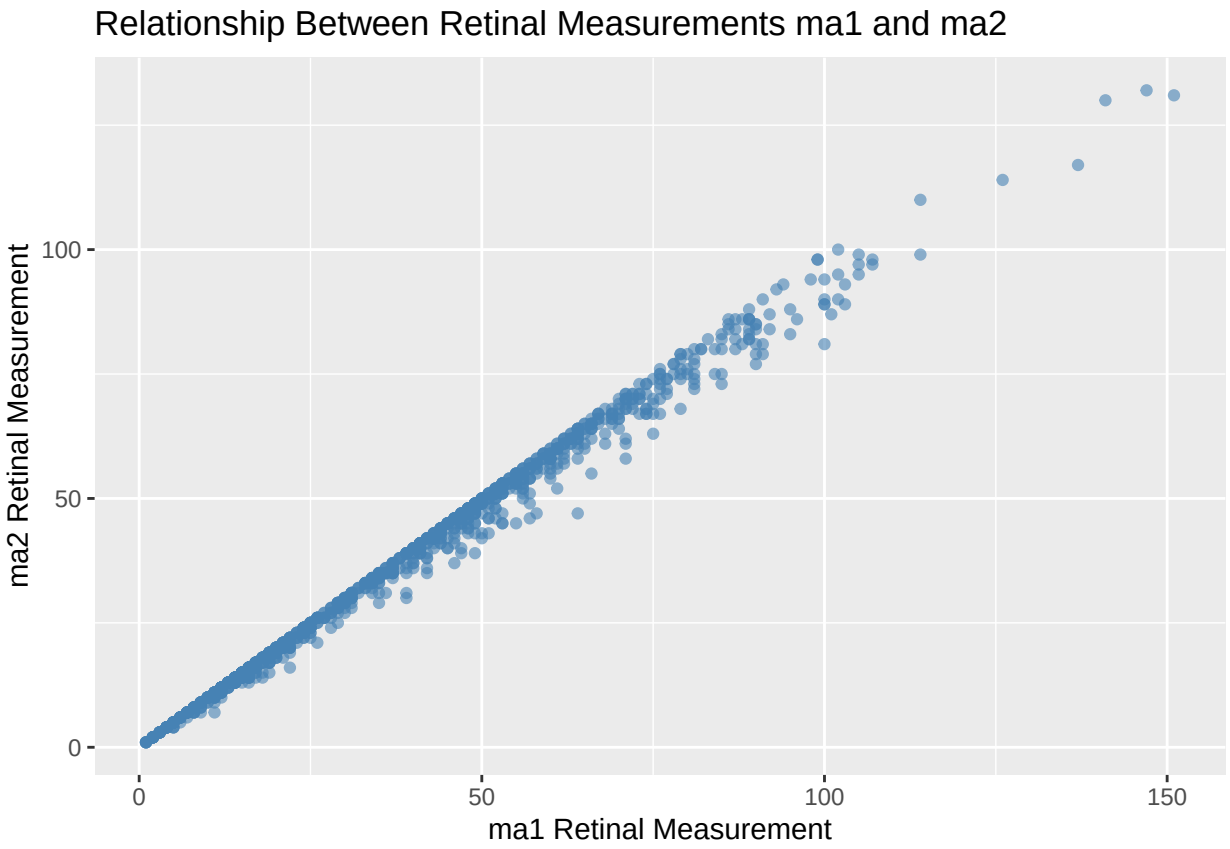
```
range_ma1
```

```
## [1]   1 151
```

14. Scatter Plot: ma1 vs ma2

```
library(ggplot2)

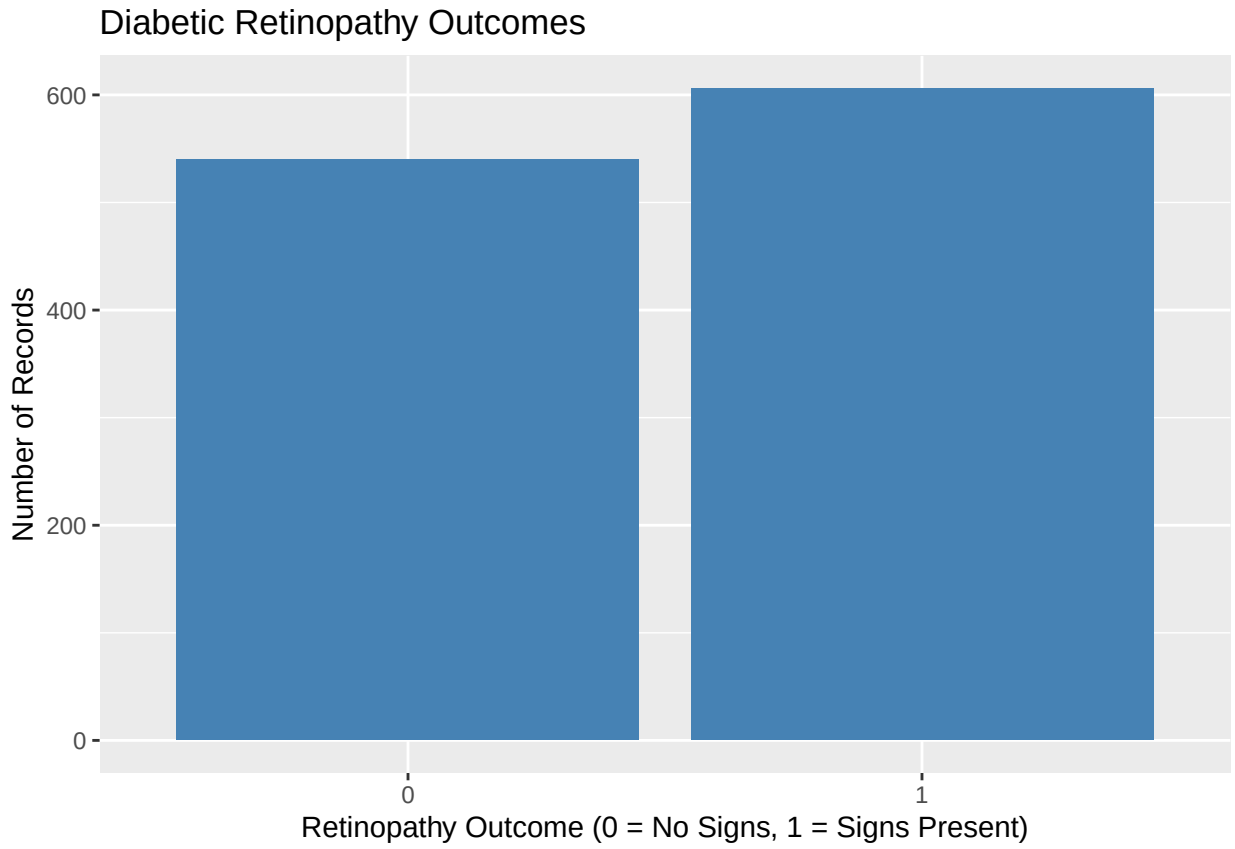
ggplot(retinopathy_clean, aes(x = ma1, y = ma2)) +
  geom_point(color = "steelblue", alpha = 0.6) +
  labs(
    title = "Relationship Between Retinal Measurements ma1 and ma2",
    x = "ma1 Retinal Measurement",
    y = "ma2 Retinal Measurement"
  )
```



15. Bar Plot: Diabetic Retinopathy Outcomes

```
# Count records by retinopathy outcome
outcome_counts <- retinopathy_clean %>%
  group_by(Retinopathy_Outcome) %>%
  summarise(Count = n())

# Create a bar plot
ggplot(outcome_counts,
       aes(x = factor(Retinopathy_Outcome),
           y = Count)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Diabetic Retinopathy Outcomes",
    x = "Retinopathy Outcome (0 = No Signs, 1 = Signs Present)",
    y = "Number of Records"
  )
```



16. Pearson Correlation Between ma1 and ma2

```
# Calculate Pearson correlation coefficient
cor(retinopathy_clean$ma1,
```

```
    retinopathy_clean$ma2,
    method = "pearson")
```

```
## [1] 0.9961732
```

```
# Perform Pearson correlation test
cor.test(retinopathy_clean$ma1,
         retinopathy_clean$ma2,
         method = "pearson")
```

```
##
##  Pearson's product-moment correlation
##
## data:  retinopathy_clean$ma1 and retinopathy_clean$ma2
## t = 385.51, df = 1144, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.9957038 0.9965914
## sample estimates:
##       cor
## 0.9961732
```